

Xinran Yu

School of Mathematical Sciences, Peking University
<https://xinrany.github.io/home/>

Employment

Peking University

Postdoctoral, Advisor: Prof. Yuguang Shi

2026–Present

Education

University of Illinois Urbana–Champaign

Ph.D. in Mathematics, Advisor: Prof. Pierre Albin

M.S. in Mathematics, May 2019

May 2026

University of Liverpool & Xi'an Jiaotong–Liverpool University

B.Sc. in Mathematics with Honors & B.Sc. in Applied Mathematics (dual degree, 2+2 program)

Thesis: *Analytic Continuation and Riemann Surfaces*

Advisor: Dr. Jon Woolf

June 2017

Preprints and Publications

1. Spectral asymmetry, supersymmetry and the equivariant Riemann-Roch defect
with Gayana Jayasinghe and Alex R. Taylor, [arXiv:2603.26057](#) 2026
2. Conformally Compact Metrics and the Lovelock Tensors 2025
[arXiv:2505.24188](#), submitted
3. Witten Instanton Complex and Morse-Bott Inequalities on Stratified Pseudomanifolds 2024
with Gayana Jayasinghe and Hadrian Quan, [arXiv:2412.12003](#), submitted

Invited Talks

1. Mini Conference on Geometric Analysis, Shanghai Jiao Tong University May 2026
2. Workshop 26w5550, Banff International Research Station Apr 2026
3. AMS Fall Central Sectional Meeting, Saint Louis University Oct 2025
4. Geometric Analysis Seminar, UIUC Sep 2025
5. Great Lakes Mathematical Physics Meeting (GLaMP), University of Kentucky Jun 2025
6. AWM Graduate Student Colloquium, UIUC Apr 2024

Honors & Awards

- Boya Postdoctoral Fellowship 2026
- Susan C. Morisato IML Scholarship, UIUC Summer 2025
- Ruth V. Shaff & Genevieve I. Andrews Fellowship, UIUC Spring 2024
- Teachers Ranked as Excellent by their Students, UIUC 2021
- Wills Prize in Mathematics, UoL 2017
- IMA Prize, UoL 2017
- University Academic Achievement Award, XJTLU 2014–2015

Teaching Experience

At UIUC

2019–2026

Instructor

- MATH 231: Calculus II (Spring 2025, Spring 2022)

Teaching Assistant

- MATH 241: Calculus III (Spring 2026, Fall 2021). Teachers Ranked as Excellent
- MATH 220: Calculus (Fall 2024)
- MATH 231: Calculus II (Spring 2021). Teachers Ranked as Excellent

Grader

Graduate Courses

- MATH 518–519: Differentiable Manifolds I–II (Fall 2023, Spring 2023)
- MATH 553: Introduction to Partial Differential Equations (Spring 2023)
- MATH 541: Functional Analysis (Spring 2023)
- MATH 512–514: Algebraic/Complex Algebraic Geometry (Fall 2022)
- MATH 525: Algebraic Topology I (Spring 2021)

Undergraduate Courses

- MATH 416: Abstract Linear Algebra (Fall 2025, Fall 2022)
- MATH 402, 403: Non-/Euclidean Geometry (Fall 2023)
- MATH 423: Differential Geometry (Fall 2023)
- MATH 286: Introduction to Differential Equations Plus (Fall 2020)
- MATH 489: Dynamics and Differential Equations (Fall 2019)
- MATH 417, 418: Introduction to Abstract Algebra I (Spring 2019), II (Fall 2019)

Mentorship

At UIUC

Illinois Mathematics Lab, Mentor

2024–2025

Projects conducted in collaboration with the Mathematics Library

- *Historical Mathematical Models* (Spring 2025) — Supervised 5 undergraduates.
 - *Documenting Historical Mathematical Models* (Fall 2024) — Supervised 4 undergraduates.
- Project received the Nancy D. Anderson Undergraduate Research Award.

Illinois Geometry Lab, Mentor

Fall 2019

- *Simulating Multi-Soliton Solutions to NLS and KdV* — Supervised 4 undergraduates.

Services & Enrichment

At UIUC

German Translation Project, Mathematics Library

2024–2025

International TA Panelist

Aug 2022

At UoL

Staff-Student Liaison Committee

2016–2017